


SOMAIYA
 VIDYAVIHAR UNIVERSITY

Semester: January 2026 – April 2026		Duration : 1hr.15min
Maximum Marks: 30		Examination: In-Semester Examination
Programme code: 16	Class: TY	Semester: VI (SVU2023)
Programme: CCE		
Name of the School: K. J. Somaiya School of Engineering		Name of the department: EXTC
Course Code: 216U41E613	Name of the Course: Machine Learning	
Note: 1. Assume suitable data where required		

Question No.		Max. Marks																						
Q1	What assumption does linear regression make about the relationship between variables? Consider below example where the mass y (grams) of a chemical is related to the time, x (seconds), for which the chemical reaction has been taking place according to the table. Find the equation of the regression line using simple linear regression. <table><tr><td>Time x(seconds)</td><td>5</td><td>7</td><td>12</td><td>16</td><td>20</td></tr><tr><td>Mass y(grams)</td><td>40</td><td>120</td><td>180</td><td>210</td><td>240</td></tr></table> OR Given a set of training data points (X,Y) apply Gradient Descent to estimate the parameters of simple linear regression model of form $\hat{Y} = \theta_0 + \theta_1 \cdot X$ Initial values of parameter are $\theta_0 = 0.1$, $\theta_1 = 0.1$ and learning rate is 0.01 <table><tr><td>X</td><td>Y</td></tr><tr><td>0.20</td><td>0.44</td></tr><tr><td>0.31</td><td>0.123</td></tr><tr><td>0.45</td><td>0.75</td></tr><tr><td>0.26</td><td>0.39</td></tr></table> a) For the given values compute the predicted value ($\hat{y}$) for each data points using initial values. b) Calculate the error for each observation. c) Compute the Sum of Squared Error(SSE). d) Find the gradient of SSE with respect to parameters and update the parameters using iteration use suitable gradient descent algorithm.	Time x (seconds)	5	7	12	16	20	Mass y (grams)	40	120	180	210	240	X	Y	0.20	0.44	0.31	0.123	0.45	0.75	0.26	0.39	10
Time x (seconds)	5	7	12	16	20																			
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Q2	a) Why is feature scaling important in KNN? Explain with the help of suitable example. b) Explain the steps involved in building a Classification Tree.	10																						
Q3	Attempt any two (5 marks each) a) Compare supervised and unsupervised learning b) Write a note on logistic regression c) Write a note on regularization	10																						



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Programme: CCE			
Name of the School: K. J. Somaiya School of Engineering		Name of the department: EXTC	
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Maximum Marks: 30		Semester: July 2025 – November 2025 Examination: In-Semester Examination (Mid Semester Examination)	Duration : 1 Hr. 15 mins
Programme code: 03 / 41 Programme: B.Tech EXTC / CCE		Class: TY	Semester: V (SVU 2023)
Name of the School: K. J. Somaiya School of Engineering		Name of the department: EXTC / CCE	
Course Code: 216U03C502, 216U41C502		Name of the Course: Computer Communication Networks	

Question No.		Max. Marks
Q1	Draw and explain TCP/IP model along with data encapsulation and decapsulation.	06
Q2 A	Solve any TWO: a) For the dataword = 10101, generate the CRC code word if divisor is 1011. b) A slotted ALOHA network transmits 300 bit frames using a shared channel with 400 Kbps bandwidth. Find the throughput (in percentage) and determine how many frames can survive, if the system produces 1000 Frames per second. c) Calculate the checksum if the given set of data words (in Hex) are to be transmitted: 23F4, B786, 69BD	10
Q2B	Solve any ONE: a) Explain the working of Go-Back-N protocol with suitable diagram. State the sender and receiver window size. b) Explain the procedure of CSMA/CA along with the flowchart.	07
Q3	Draw and explain IP header format.	07



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Semester: July 2025-November 2025			Duration :1H 15M
Maximum Marks:30	Examination: In-Semester Examination		
(Mid Semester Examination)			
Programme code: 03 Programme: B.Tech in Electronics and Telecommunication Engineering		Class: TY	Semester: V (SVU 2020/SVU2023/SVU2025)
Name of the School: K. J. Somaiya School of Engineering		Name of the department: EXTC	
Course Code: 216U03C503		Name of the Course: Digital Signal Processing	

Question No.		Max. Marks
Q1	Attempt any four. (5 Marks each) I) Given the transfer function: $H(z) = \frac{1+0.5Z^{-1}-0.5Z^{-2}}{1-1.3Z^{-1}+0.36Z^{-2}}$ Find the cascade realization of the filter and describe the block diagram. II) Perform circular convolution of the following sequences. $p(n) = \{1, 3, 1, 3, 1, 3\}$ and $q(n) = \{-1, 2, -1, 2\}$ III) Write the procedure for designing FIR filter by windows. IV) Perform crosscorrelation of the given sequences. $x(n) = \{1, 2, 1, 2\}$; $h(n) = \{2, 1, 2, 1\}$ V) For a given $h(n) = [0.5, -0.25, 1]$, Draw a pole-zero plot of the system.	20
Q2	Attempt any one (10 marks each) I) Find 8-point DFT of the discrete time signal using radix-2 DIT FFT. $x(n) = [2, 0, 1, 0, 1, 0, 1, 0]$ II) Find 8-point DFT of the discrete time signal using radix-2 DIF FFT. $x(n) = [2, 0, 1, 0, 1, 0, 1, 0]$	10



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TY - EXTC

Maximum Marks: 30

Semester: July 2025 - November 2025
Examination: In-Semester Examination

Duration: 1hr. 15 min.

Programme code: 27

Programme: B.Tech., Honors- Data Science for
Emerging Analytics, (CCE, EXTC)

Class: TY

Semester: IV
(SVU 2023)

Name of the Constituent College:

K. J. Somaiya School of Engineering

Name of the department:
EXTC

Course Code: 216h27C501

Name of the Course: Data Mining & Warehousing

Note: 1. Assume suitable data where required

Question No.		Max. Marks																																																				
Q. 1	Answer <u>ANY TWO</u> of the following: A. How does Data Mining work as a step in the process of Knowledge Discovery? Explain using diagram. B. Summarize about clustering. Describe different clustering methods with suitable examples. C. Describe classifier performance evaluation metrics with relevant formulae. State with justification which metrics are appropriate for imbalanced data. D. Describe the concept of ‘bias-variance tradeoff’ in relation to the selection of a generalized machine learning model for data mining purpose. E. What is ensemble learning, and how does it differ from using a single machine learning model? Describe ADABOOST algorithm in detail.	20																																																				
Q. 2	Answer <u>ANY ONE</u> of the following: A database has the following five transactions. Let min_support = 80% and min_confidence = 20%. Find the frequent item sets using Apriori algorithm. <table><tr><th>T id</th><th>Items bought</th></tr><tr><td>T1</td><td>K, A, D, B</td></tr><tr><td>T2</td><td>D, A, C, E, B</td></tr><tr><td>T3</td><td>C, A, E, B</td></tr><tr><td>T4</td><td>B, A, D</td></tr></table> OR A Healthcare institute wants to build a model to predict the risk of “Heart attack” based on the selected predictors shown in the following table. Use ID3 algorithm to train the model. Sketch the resultant tree and show all the calculations. <table><tr><th>Patient ID</th><th>Chest pain?</th><th>Male?</th><th>Smokes?</th><th>Exercises?</th><th>Heart attack?</th></tr><tr><td>1</td><td>Yes</td><td>Yes</td><td>No</td><td>Yes</td><td>Yes</td></tr><tr><td>2</td><td>Yes</td><td>Yes</td><td>Yes</td><td>No</td><td>Yes</td></tr><tr><td>3</td><td>No</td><td>No</td><td>Yes</td><td>No</td><td>Yes</td></tr><tr><td>4</td><td>No</td><td>Yes</td><td>No</td><td>Yes</td><td>No</td></tr><tr><td>5</td><td>Yes</td><td>No</td><td>Yes</td><td>Yes</td><td>Yes</td></tr><tr><td>6</td><td>No</td><td>Yes</td><td>Yes</td><td>Yes</td><td>No</td></tr></table>	T id	Items bought	T1	K, A, D, B	T2	D, A, C, E, B	T3	C, A, E, B	T4	B, A, D	Patient ID	Chest pain?	Male?	Smokes?	Exercises?	Heart attack?	1	Yes	Yes	No	Yes	Yes	2	Yes	Yes	Yes	No	Yes	3	No	No	Yes	No	Yes	4	No	Yes	No	Yes	No	5	Yes	No	Yes	Yes	Yes	6	No	Yes	Yes	Yes	No	10
T id	Items bought																																																					
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3	No	No	Yes	No	Yes																																																	
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6	No	Yes	Yes	Yes	No																																																	



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Semester: July 2025 - November 2025		
Maximum Marks: 30	Examination: In-Semester Examination	Duration: 1hr. 15 min.
Programme code: 03		
Programme: B.Tech , EXTC	Class: TY	Semester: V (SVU 2023)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: EXTC
Course Code: 216U03E515	Name of the Course: Machine Learning	
Note: 1. Assume suitable data where required		

Question No.		Max. Marks																					
Q. 1	Answer any TWO of the following(Each question carries 10 marks) A. Differentiate between supervised, unsupervised and reinforcement learning methods. Differentiate between classification, regression and clustering tasks with one example of each. B. What is machine learning? Describe the steps involved in building a machine learning model. State one example where you apply machine learning and where you cannot apply machine learning. C. Explain k-nearest neighbor algorithm. State its merits and drawbacks. D. Describe Perceptron along with learning algorithm. What are its drawbacks? How they are overcome by MLP?	20																					
Q. 2	In this question we investigate whether students will Pass or Fail Machine Learning Examination based on GPA in previous semester(Low, Medium, High) and whether or not they studied(Yes, No) well before the exam. You are given the following data for six students. There are two features, "GPA" and "Studied" The column "Result" shows the label we want to predict. Use <u>EITHER</u> Decision Tree(ID3) algorithm <u>OR</u> Naïve Bayes' algorithm to build a model for binary classification using following data. <table><tr><td>GPA</td><td>Studied</td><td>Result</td></tr><tr><td>Low</td><td>Yes</td><td>Pass</td></tr><tr><td>Low</td><td>No</td><td>Fail</td></tr><tr><td>Medium</td><td>Yes</td><td>Pass</td></tr><tr><td>Medium</td><td>No</td><td>Fail</td></tr><tr><td>High</td><td>Yes</td><td>Pass</td></tr><tr><td>High</td><td>No</td><td>Pass</td></tr></table>	GPA	Studied	Result	Low	Yes	Pass	Low	No	Fail	Medium	Yes	Pass	Medium	No	Fail	High	Yes	Pass	High	No	Pass	10
GPA	Studied	Result																					
Low	Yes	Pass																					
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Maximum Marks:30		Semester: July 2025-November 2025		Duration :1H 15M
		Examination: In-Semester Examination (Mid Semester Examination)		
Programme code: 03		Class: TY	Semester: V	
Programme: B.Tech in Electronics and Telecommunication Engineering			(SVU) 2020/SVU2023/SVU2025	
Name of the School:		Name of the department:		
K. J. Somaiya School of Engineering		EXTC		
Course Code: 216U03C503		Name of the Course: Digital Signal Processing		

Question No.		Max. Marks
Q1	Attempt any four. (5 Marks each) I) Given the transfer function: $H(z) = \frac{1+0.5Z^{-1}-0.5Z^{-2}}{1-1.3Z^{-1}+0.36Z^{-2}}$ Find the cascade realization of the filter and describe the block diagram. II) Perform circular convolution of the following sequences. $p(n) = \{1, 3, 1, 3, 1, 3\}$ and $q(n) = \{-1, 2, -1, 2\}$ III) Write the procedure for designing FIR filter by windows. IV) Perform crosscorrelation of the given sequences. $x(n) = \{1, 2, 1, 2\}$; $h(n) = \{2, 1, 2, 1\}$ V) For a given $h(n) = [0.5, -0.25, 1]$, Draw a pole-zero plot of the system.	20
Q2	Attempt any one (10 marks each) I) Find 8-point DFT of the discrete time signal using radix-2 DIT FFT. $x(n) = [2, 0, 1, 0, 1, 0, 1, 0]$ II) Find 8-point DFT of the discrete time signal using radix-2 DIF FFT. $x(n) = [2, 0, 1, 0, 1, 0, 1, 0]$	10

**SOM**

Maximum Marks: 30		Semester: July 2025-November 2025		Duration: 1 hr. 15 min
Examination: In-Semester Examination (Mid Semester Examination)				
Programme code: 03		Class: TY		Semester: V (SVU2023)
Programme: Electronics and Telecommunication Engineering				
Name of the School:		Name of the department:		
K. J. Somaiya School of Engineering		EXTC		
Course Code: 216U03C501		Name of the Course: Digital Communication		

Question No.		Max. Marks																
Q1	Find capacity of Gaussian channel of bandwidth 4KHz and noise PSD 10^{-9} W/Hz when signal energy is a) 0.1J b)how does channel capacity changes in b if bandwidth is increased to 10KHz?	5																
A																		
B	The signal $x(t) = 20 + 20 \sin(500\pi t + 30^\circ)$ is to be sampled periodically and reproduced from the samples 1. Find the maximum allowable time interval between the samples 2. How many sample values need to be stored in order to reproduce 1 sec of this waveform, if sampled according to the result in (1) 3. Determine and sketch spectrum of sampled signal when sampling frequency $f_s = 750$ Hz OR Define line coding. Draw the waveforms for a binary sequence 10110011 for following signal codes : (1) Polar R Z (2) Unipolar NRZ (3) Manchester (4) Alternate Mark Invasion (AMI)	10																
Q2	A discrete memoryless source has alphabet of five symbols with probability for its output as given below: <table border="1"><thead><tr><th>Symbol</th><th>S_0</th><th>S_1</th><th>S_2</th><th>S_3</th><th>S_4</th><th>S_5</th><th>S_6</th></tr></thead><tbody><tr><td>Probability</td><td>0.1</td><td>0.2</td><td>0.3</td><td>0.04</td><td>0.005</td><td>0.005</td><td>0.35</td></tr></tbody></table> (i) Design the HUFFmann code for the above symbols (ii) Find Entropy (iii) Compute average codeword length (iv) Calculate Code Efficiency	Symbol	S_0	S_1	S_2	S_3	S_4	S_5	S_6	Probability	0.1	0.2	0.3	0.04	0.005	0.005	0.35	10
Symbol	S_0	S_1	S_2	S_3	S_4	S_5	S_6											
Probability	0.1	0.2	0.3	0.04	0.005	0.005	0.35											

Q3	State the Hartley Shannon theorem expressing Channel capacity OR What is entropy? How is it related to information?	5
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Semester: July 2025 – November 2025

Maximum Marks: 30

**Examination: In-Semester Examination
(Mid Semester Examination)**

Duration : 1 Hr. 15 mins

Programme code: 03 / 41

Programme: B.Tech EXTC / CCE

Class: TY

**Semester: V
(SVU 2023)**

Name of the School:

K. J. Somaiya School of Engineering

Name of the department:

EXTC / CCE

**Course Code: 216U03C502,
216U41C502**

Name of the Course: Computer Communication Networks

Question No.		Max. Marks
Q1	Draw and explain TCP/IP model along with data encapsulation and decapsulation.	06
Q2 A	Solve any TWO: a) For the dataword = 10101, generate the CRC code word if divisor is 1011. b) A slotted ALOHA network transmits 300 bit frames using a shared channel with 400 Kbps bandwidth. Find the throughput (in percentage) and determine how many frames can survive, if the system produces 1000 Frames per second. c) Calculate the checksum if the given set of data words (in Hex) are to be transmitted: 23F4, B786, 69BD	10
Q2B	Solve any ONE: a) Explain the working of Go-Back-N protocol with suitable diagram. State the sender and receiver window size. b) Explain the procedure of CSMA/CA along with the flowchart.	07
Q3	Draw and explain IP header format.	07



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Maximum Marks: 30		Semester: July 2025-November 2025		Duration :1hr 15mins	
Examination: In-Semester Examination (Mid Semester Examination)					
Programme code: 03		Class: TY		Semester: V	
Programme: Electronics and Telecommunication				(SVU :SVU2023)	
Name of the School: K. J. Somaiya School of Engineering			Name of the department: EXTC		
Course Code: 216U03E513			Name of the Course: Database Management System		

Question No.		Max. Marks																																			
Q1	A company maintains the details of its hardware suppliers in a database. The table Suppliers is created with the following fields: <table><tr><th>Sup_No (Primary Key)</th><th>Sup_Name</th><th>Item_Supplied</th><th>Item_Price</th><th>City</th></tr><tr><td>S11</td><td>Anita</td><td>Monitor</td><td>12000</td><td>Pune</td></tr><tr><td>S12</td><td>Rahul</td><td>Hard Disk</td><td>5000</td><td>Kolkata</td></tr><tr><td>S13</td><td>Rekha</td><td>RAM</td><td>3000</td><td>Delhi</td></tr><tr><td>S14</td><td>Sunil</td><td>Hard Disk</td><td>5500</td><td>Hyderabad</td></tr><tr><td>S15</td><td>Ritu</td><td>Monitor</td><td>11500</td><td>Bangalore</td></tr><tr><td>S16</td><td>Ajay</td><td>RAM</td><td>3200</td><td>Mumbai</td></tr></table> 1. Write an SQL query to display the names of suppliers who supply Hard Disk and are located in Hyderabad. 2. Write an SQL query to display the supplier name and item with the maximum price. 3. Write an SQL query to count the number of suppliers from Delhi and Mumbai. 4. Write an SQL query to display all suppliers in ascending order of Item Price.	Sup_No (Primary Key)	Sup_Name	Item_Supplied	Item_Price	City	S11	Anita	Monitor	12000	Pune	S12	Rahul	Hard Disk	5000	Kolkata	S13	Rekha	RAM	3000	Delhi	S14	Sunil	Hard Disk	5500	Hyderabad	S15	Ritu	Monitor	11500	Bangalore	S16	Ajay	RAM	3200	Mumbai	10 Marks
Sup_No (Primary Key)	Sup_Name	Item_Supplied	Item_Price	City																																	
S11	Anita	Monitor	12000	Pune																																	
S12	Rahul	Hard Disk	5000	Kolkata																																	
S13	Rekha	RAM	3000	Delhi																																	
S14	Sunil	Hard Disk	5500	Hyderabad																																	
S15	Ritu	Monitor	11500	Bangalore																																	
S16	Ajay	RAM	3200	Mumbai																																	
Q2	Answer any two: a. What are the objectives of Normalization? b. Compare ER and EER c. Differentiate between DBMS and RDBMS. Give at least four points of comparison.	5 Marks 5 Marks																																			
Q3	Answer any one: a. Explain Triggers in SQL with example b. Explain the differences between Primary Key, Foreign Key, and Unique Key with suitable examples.	10 Marks																																			



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Maximum Marks: 30		Semester: July 2025-November 2025	
		Examination: In-Semester Examination Duration :1.hr 15 mins	
		(Mid Semester Examination)	
Programme code: 03			
Programme: Electronics and Telecommunication Engineering		Class: LY	Semester: VII (SVU2020)
Name of the School:		Name of the department:EXTC	
K. J. Somaiya School of Engineering			
Course Code: 116U03E721		Name of the Course: Real Time Operating System	

Question No.		Max. Marks
Q1	Compare medium scale and sophisticated embedded systems. What is the significance of programming model? Enlist the types of programming models.	10
Q2	What is the difference between hard and soft real time systems? Explain Rate-monotonic scheduling with suitable real time example. OR What is the job of task scheduler? Discuss performance metric in scheduling models.	10
Q3	Solve any two I. Write a note on I/O management in RTOS II. Explain any three design metrics of embedded systems III. Write a note on binary semaphore	10

SY - CCE

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Maximum Marks: 30	Semester: January 2025 - April 2025	Duration: 1hr. 15 min.
Programme code: 41	Examination: In-Semester Examination	
Programme: B.Tech Computer and Communication Engineering	Class: SY	Semester: IV (SVU 2023)
Institute/school/Department	Name of the department:	
K. J. Somaiya School of Engineering	EXTC	
Course Code: 216U41C405	Name of the Course: Operating System	

Question No.		Max. Marks																				
Q1	Write short notes on (Any 2) :- 1) Compare medium term and long term schedulers. 2) Explain any 2 functions of OS in short. 3) Draw and explain five state process diagram	10 (5 marks Each)																				
Q2 (A)	What is "Race condition". Explain with example.	5																				
Q2 (B)	Discuss how deadlock could be detected.	5																				
Q3	What is a semaphore? Explain types of semaphores with appropriate example. OR Consider the following snapshot of the process to be executed. Draw the Gantt chart and determine the average waiting time and average turnaround time for FCFS, SJF (non-preemptive), and priority (preemptive) scheduling algorithm. Note: Assume lower number indicates higher priority <table><tr><th>Process</th><th>Arrival Time</th><th>Burst Time</th><th>Priority</th></tr><tr><td>P1</td><td>0</td><td>7</td><td>2</td></tr><tr><td>P2</td><td>1</td><td>6</td><td>1</td></tr><tr><td>P3</td><td>3</td><td>4</td><td>2</td></tr><tr><td>P4</td><td>5</td><td>2</td><td>3</td></tr></table>	Process	Arrival Time	Burst Time	Priority	P1	0	7	2	P2	1	6	1	P3	3	4	2	P4	5	2	3	10
Process	Arrival Time	Burst Time	Priority																			
P1	0	7	2																			
P2	1	6	1																			
P3	3	4	2																			
P4	5	2	3																			



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Maximum Marks: 30	Semester: January 2025 - April 2025	Duration :1 hr.15 min.
Programme code: 03	Examination: In-Semester Examination	
Programme: B.Tech Electronics and Telecommunication Engineering	Class: TY	Semester: VI (SVU 2020)
Institute/School/ Department: K. J. Somaiya School of Engineering	Name of the department: EXTC	
Course Code: 116U03C602	Name of the Course: Computer Communication Networks	

Question No.		Max. Marks
1	Explain OSI model along with data encapsulation and decapsulation. OR Explain the TCP/IP model and discuss various types of addresses used at different layers and their uses.	06
2	A) Explain the working of slotted ALOHA protocol with suitable diagram and state it's vulnerable period and maximum throughput. In a pure ALOHA network transmits 200 bits frame on a channel of 400 Kbps. What is the requirement to make this frame collision-free?	07
	B) How Address resolution protocol works, explain with network diagram ? Draw and explain ARP packet format	07
3	A) Identify the class of each IP address. Determine Network IP address and Broadcast IP address for the following IP addresses : i) 123.28.115.0 ii) 92.215.3.15/24 iii) 183.68.15.9	05
	B) Explain the following fields w.r.t. the IP header : i) Total Length ii) Identification iii) Flags iv) TTL v) HLEN OR B) What is the need of fragmentation in IP ? explain with suitable example	05



SOMAIYA
VIDYAVIHAR UNIVERSITY

Semester: July 2024-November 2024			Duration: 1hr. 15 min.
Maximum Marks: 30	Examination: In-Semester Examination		
Programme code: 03	Class: TY	Semester: V (SVU 2020/SVU 2023)	
Programme: B.Tech Electronics and Communication Engineering			
Name of the Constituent College: Somaiya College of Engineering		Name of the department: EXTC	
Course Code: 116U03C503	Name of the Course: Digital Signal Processing		

Question No.		Max. Marks
1	Determine the 8 point DFT of $x(n)$ using Radix-2 DITFFT algorithm also draw the flow graph for the same. $X(n)=\{1, 2, 1, 2, 3, 1, 3, 1\}$ OR a) State and prove the linearity property of DFT. b) State and prove the time reversal property of DFT.	10 5 5
2	Attempt any Two. (a) Perform the circular convolution of $x_1(n)=\{1, 2, 3, 4\}$ $x_2(n)=\{4, 3, 2, 1\}$ (b) Find the autocorrelation of the sequence for $x(n)=\{1, 2, 3, 4\}$ (c) Perform the linear convolution of the following sequences by overlap and add method. $x_1(n)=\{1, 2, 3, 3, 2, 1, 2, 3, 4, 2\}$, $x_2(n)=\{1, 2, 1\}$	10
3	Obtain the Direct form –I and Direct form – II form of realization of the LTI system governed by the equation $Y(n) = 4 y(n-3) + 3 y(n-2) - 12 y(n-1) + x(n) + 0.5 x(n-1) - 0.75 x(n-2)$	10



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VIDYAVIHAR UNIVERSITY

Semester: July 2024-November 2024		Duration: 1hr. 15 min.
Num Marks: 30	Examination: In-Semester Examination	
Programme code: 03	Class: TY	Semester: V (SVU 2020/SVU 2023)
Programme: B.Tech Electronics and Communication Engineering		
Name of the Constituent College: Somaia College of Engineering		Name of the department: EXTC
Course Code: 116U03C503	Name of the Course: Digital Signal Processing	

Question		Max. Marks
1	Determine the 8 point DFT of $x(n)$ using Radix-2 DITFFT algorithm also draw the flow graph for the same. $X(n) = \{1, 2, 1, 2, 3, 1, 3, 1\}$ OR a) State and prove the linearity property of DFT. b) State and prove the time reversal property of DFT.	10 5 5
2	Attempt any Two. (a) Perform the circular convolution of $x_1(n) = \{1, 2, 3, 4\}$ $x_2(n) = \{4, 3, 2, 1\}$ (b) Find the autocorrelation of the sequence for $x(n) = \{1, 2, 3, 4\}$ (c) Perform the linear convolution of the following sequences by overlap and add method. $x_1(n) = \{1, 2, 3, 3, 2, 1, 2, 3, 4, 2\}$, $x_2(n) = \{1, 2, 1\}$	10
3	Obtain the Direct form –I and Direct form – II form of realization of the LTI system governed by the equation $Y(n) = 4y(n-3) + 3y(n-2) - 12y(n-1) + x(n) + 0.5x(n-1) - 0.75x(n-2)$	10



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VIDYAVIHAR UNIVERSITY

Semester: July 2024-November 2024		
Num Marks: 30	Examination: In-Semester Examination	Duration: 1hr. 15 min.
Programme code: 03	Class: TY	Semester: V (SVU 2020/SVU 2023)
Programme: B.Tech Electronics and Communication Engineering		
Name of the Constituent College: Somaiya College of Engineering		Name of the department: EXTC
Course Code: 116U03C503	Name of the Course: Digital Signal Processing	

Q. No.	Question	Max. Marks
1	Determine the 8 point DFT of $x(n)$ using Radix-2 DITFFT algorithm also draw the flow graph for the same. $X(n) = \{1, 2, 1, 2, 3, 1, 3, 1\}$ OR a) State and prove the linearity property of DFT. b) State and prove the time reversal property of DFT.	10 5 5
2	Attempt any Two. (a) Perform the circular convolution of $x_1(n) = \{1, 2, 3, 4\}$ $x_2(n) = \{4, 3, 2, 1\}$ (b) Find the autocorrelation of the sequence for $x(n) = \{1, 2, 3, 4\}$ (c) Perform the linear convolution of the following sequences by overlap and add method. $x_1(n) = \{1, 2, 3, 3, 2, 1, 2, 3, 4, 2\}$, $x_2(n) = \{1, 2, 1\}$	10
3	Obtain the Direct form -I and Direct form - II form of realization of the LTI system governed by the equation $Y(n) = 4y(n-3) + 3y(n-2) - 12y(n-1) + x(n) + 0.5x(n-1) - 0.75x(n-2)$	10



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VIDYAVIHAR UNIVERSITY

Semester: July 2024 - November 2024		
Maximum Marks: 30	Examination: In-Semester Examination	Duration :1 hr.15 min.
Programme code: 03		
Programme: B.Tech Electronics and Communication Engineering	Class: TY/EXTC	Semester: V (SVU 2023)
Name of the Constituent College: Somaiya College of Engineering		Name of the department: EXTC
Registration Code: 116U03E513	Name of the Course: Operating System	

Questions	Max. Marks
Write short notes on (Any 2) :- 1) System calls. 2) Explain any 2 functions of OS in short. 3) Comparison of Monolithic and Micro kernel.	10
What are the four characteristics of deadlock to occur?	05
Explain two process Peterson's solution to achieve mutual exclusion of critical sections in an OS with proper code and example.	05
What is semaphores? Explain types of semaphores with appropriate example. OR Consider the following snapshot of the process to be executed. Draw the Gantt chart and determine the average waiting time and average turnaround time for FCFS, SJF (non-preemptive), and priority(preemptive) scheduling algorithm. Note: Assume lower number indicates higher priority	10

Process	Arrival Time	Burst Time	Priority
P1	0	7	2
P2	1	5	1
P3	3	4	2
P4	5	3	3

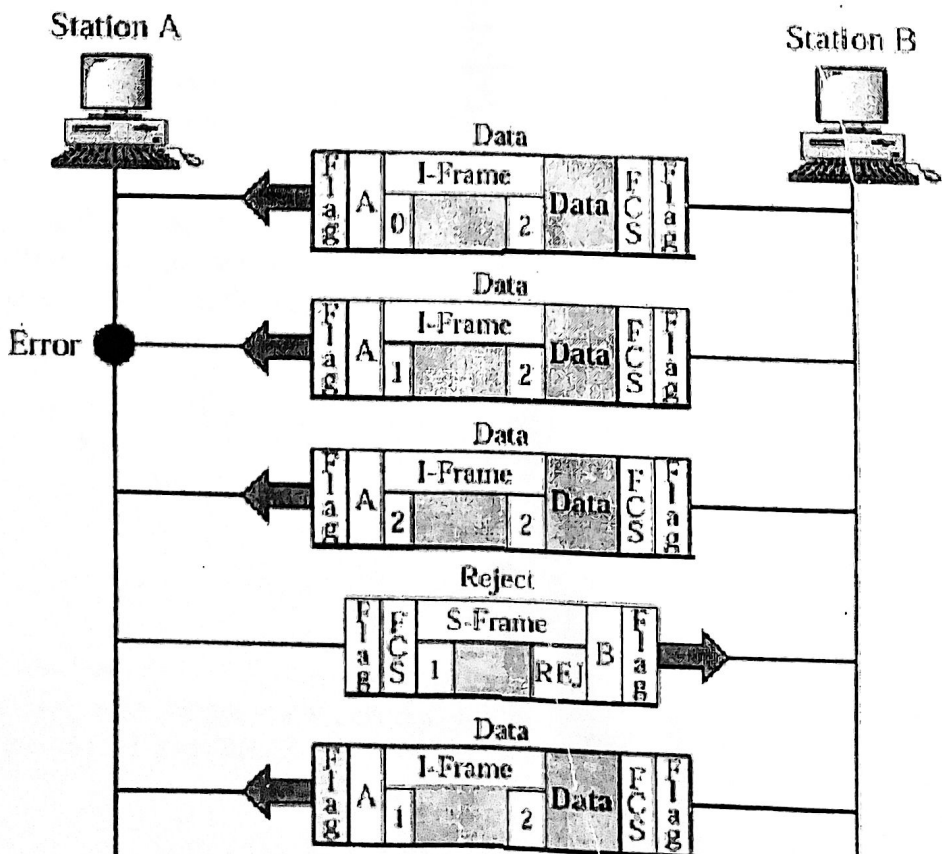


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Semester: January 2024 – April 2024		
Maximum Marks: 30	Examination: In-Semester Examination	Duration :1 hr.15 min.
Programme code: 03(UG) Programme: B.Tech Electronics and Communication Engineering	Class:SY	Semester: IV (SVU 2020)
Name of the Constituent College: Somaiya College of Engineering		Name of the department: EXTC
Course Code: 116U06G601	Name of the Course: Broadband Communications (Honors- Advanced Communications)	

	Max. Marks
Explain in brief (any 2): 1. Advantages of OSI model 2. Causes of network congestion in broadband networks 3. Data link layer protocols	10

Semester: January 2024- April 2024			
Maximum Marks:30		Examination: In-Semester Examination	
		Duration :1hr15min	
Programme code: B.Tech		Class: TY	Semester: VI (SVU 2020)
Programme: 03			
Name of the Constituent College:		Name of the department: EXTC	
S. Somaiya College of Engineering			
Course Code: 116U03C602		Name of the Course: Computer Communication Networks	

	Max. Marks
With respect to OSI model, explain the communication data flow between the two hosts via intermediate routers , with the help of neat diagram.	05 marks
With the help of flowchart, briefly explain the procedure for CSMA/CD. A CSMA/CD network has a bandwidth of 100 Mbps. If the maximum propagation delay is $51.2 \mu\text{s}$, determine the minimum frame size . <u>OR</u> State the use of S-frame ,I-frame and U-frame of HDLC protocol. Interpret the diagram given below and answer the following questions : i) How many frames B has successfully sent to A ? ii) How many information frames station B has already received from A ? iii) What do top 3 frames indicate? (iv) What do last 2 frames indicate?	10marks (5+5)
	10marks (2+8)

Q2 (B)	Calculate the checksum if the given set of data words (in Hex) are to be transmitted: 23F4 , B786 , 69BD	05 marks
Q3	Solve any TWO: a) Draw and briefly explain IPv4 header format. b) State the class of each address, Network IP address and Broadcast IP address, or the following IP addresses : i) 165.122.35.40 ii) 214.255.45.69 iii) 112.225.3.215 c) If the IPv4 packet is arrived with the first few Hex digits as 48000028000100000602 determine: i) the length of header in bytes, ii) length of the data bytes in IP datagram and iii) maximum number of hops the IP packet is allowed.	10 marks (5 marks each)



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VIDYAVIHAR UNIVERSITY

Semester: January 2024- April 2024		
Maximum Marks: 30	Examination: In-Semester Examination	Duration :1.15 hours
Programme code: 03	Class: TY	Semester: VI
Programme: B. Tech.		(SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: EXTC
Course Code: 116U03E613	Name of the Course: Machine Learning	

Instructions: 1. Your answers should show all calculations for full marks
2. Please be precise in your answers

Question No.		Max. Marks																																																		
Q. 1.	Answer any FOUR questions from following(each question carries FIVE marks) A. Following table shows result of binary classification task performed on a test data set. <table><tr><td>Actual label</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>Predicted label</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td></tr></table> Build confusion matrix. Compute accuracy, precision, recall and f-score for class “1”. B. Using the dataset given below, build a decision tree which classifies Y as T/F given the binary predictors A, B, C. Use ID3 algorithm and draw the learned tree. Does this tree give zero training error? Show all computations. <table><tr><td>A</td><td>B</td><td>C</td><td>Y</td></tr><tr><td>F</td><td>F</td><td>F</td><td>F</td></tr><tr><td>T</td><td>F</td><td>T</td><td>T</td></tr><tr><td>T</td><td>T</td><td>F</td><td>T</td></tr><tr><td>T</td><td>T</td><td>T</td><td>F</td></tr></table> C. A machine learning algorithm has 3 hyper-parameters with h_1, h_2, h_3 number of values respectively. How many models will be built using 5-fold cross-validation to choose the best model? Explain. D. State the requirement for data scaling in some Machine Learning algorithms. Consider the dataset given below. Transform the weights to range [40, 60] using Max-Min scaling method. <table><tr><td>Name of candidate</td><td>Rahul</td><td>Simran</td><td>Jay</td><td>Punita</td><td>Sara</td></tr><tr><td>Weight</td><td>70</td><td>65</td><td>80</td><td>48</td><td>55</td></tr></table>	Actual label	1	0	1	0	0	1	1	1	Predicted label	0	0	1	1	0	1	0	1	A	B	C	Y	F	F	F	F	T	F	T	T	T	T	F	T	T	T	T	F	Name of candidate	Rahul	Simran	Jay	Punita	Sara	Weight	70	65	80	48	55	20
Actual label	1	0	1	0	0	1	1	1																																												
Predicted label	0	0	1	1	0	1	0	1																																												
A	B	C	Y																																																	
F	F	F	F																																																	
T	F	T	T																																																	
T	T	F	T																																																	
T	T	T	F																																																	
Name of candidate	Rahul	Simran	Jay	Punita	Sara																																															
Weight	70	65	80	48	55																																															

E. Can the learning rate for gradient descent algorithm be chosen of any random value? State the consequences of choosing any random value for the learning rate with the help of a relevant sketch.

F. State the steps in k-nearest neighbour algorithm. Following figure illustrates decision boundaries for two nearest-neighbour classifiers. Determine which one of the boundaries belongs to the 1-nearest neighbour classifier and which one belongs to the 3-nearest neighbour classifier.

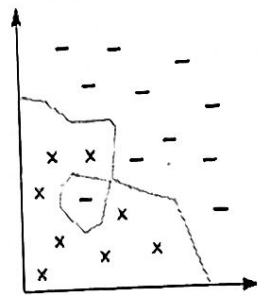


Fig.1 Decision Boundary 1

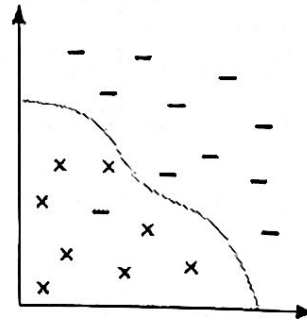


Fig.2 Decision Boundary 2

Q.2. Overfitting is a commonly occurring phenomenon in Machine Learning models. State the common reasons for overfitting. Describe how to identify if the model is overfitting to train data. State and explain any one technique to minimize overfitting.

OR

State the difference between following Machine Learning algorithms in terms of given criteria:

- A. Simple linear regression
- B. Multiple linear regression
- C. Polynomial regression
- D. Logistic regression

Criteria: 1. Hypothesis function, 2. Cost function, 3. Learning algorithm, 4. Application, 5. Model parameters



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VIDYAVIHAR UNIVERSITY

Semester: July 2023 – Oct 2023		
Maximum Marks: 30	Examination: In-Semester Examination	Duration :1 hr.15 min.
Programme code: 03		
Programme: B.Tech Electronics and Telecommunication Engineering	Class: TY	Semester: V (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: EXTC	
Course Code: 116U03C501	Name of the Course: Digital Communications	

Question No.		Max. Marks	CO Mapped														
Q1	Write a short note on any two A. Gram Schmidt orthonormalization with reference to signal representation B. Channel Capacity Theorem C. Sampling theorem for band pass signal	10	CO1, CO2, CO3														
Q2	A discrete memoryless source has alphabet of six symbols with probability for its output as given below: <table border="1"><tr><td>Symbol</td><td>S₀</td><td>S₁</td><td>S₂</td><td>S₃</td><td>S₄</td><td>S₅</td></tr><tr><td>Probability</td><td>0.1</td><td>0.05</td><td>0.2</td><td>0.2</td><td>0.15</td><td>0.3</td></tr></table> (i) Design the Shannon-Fano code for the above symbols (ii) Find Entropy (iii) Compute average codeword length (iv) Calculate Code Efficiency OR Encode the following source string using Lempel-Ziv algorithm b c d a c d b a a b d c	Symbol	S ₀	S ₁	S ₂	S ₃	S ₄	S ₅	Probability	0.1	0.05	0.2	0.2	0.15	0.3	10	CO2
Symbol	S ₀	S ₁	S ₂	S ₃	S ₄	S ₅											
Probability	0.1	0.05	0.2	0.2	0.15	0.3											
Q3	A. Draw polar NRZ and Manchester encoding waveform for following data sequence. 1 1 0 0 1 0 1 1 0 Comment on power Spectral density of both codes with diagram. B. X is an unknown 8-bit binary number. You are given another 8-bit binary number, Y, and told that Y differs from X in exactly one bit position. How many bits of information about X have you been given?	6+4	CO2, CO3														



SOMAIYA
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Maximum Marks: 30	Semester: July 2023 – Oct 2023	Examination: In-Semester Examination	Duration :1 hr.15 min.
Programme code: 03			
Programme: B.Tech Electronics and Telecommunication Engineering	Class: TY	Semester: V (SVU 2020)	
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: EXTC	
Course Code: 116U03C501	Name of the Course: Digital Communications		

Question No.		Max. Marks	CO Mapped														
Q1	Write a short note on any two A. Gram Schmidt orthonormalization with reference to signal representation B. Channel Capacity Theorem C. Sampling theorem for band pass signal	10	CO1, CO2, CO3														
Q2	A discrete memoryless source has alphabet of six symbols with probability for its output as given below: <table border="1"><tr><td>Symbol</td><td>S₀</td><td>S₁</td><td>S₂</td><td>S₃</td><td>S₄</td><td>S₅</td></tr><tr><td>Probability</td><td>0.1</td><td>0.05</td><td>0.2</td><td>0.2</td><td>0.15</td><td>0.3</td></tr></table> (i) Design the Shannon-Fano code for the above symbols (ii) Find Entropy (iii) Compute average codeword length (iv) Calculate Code Efficiency OR Encode the following source string using Lempel-Ziv algorithm b c d a c d b a a b d c	Symbol	S ₀	S ₁	S ₂	S ₃	S ₄	S ₅	Probability	0.1	0.05	0.2	0.2	0.15	0.3	10	CO2
Symbol	S ₀	S ₁	S ₂	S ₃	S ₄	S ₅											
Probability	0.1	0.05	0.2	0.2	0.15	0.3											
Q3	A. Draw polar NRZ and Manchester encoding waveform for following data sequence. 1 1 0 0 1 0 1 1 0 Comment on power Spectral density of both codes with d'igram. B. X is an unknown 8-bit binary number. You are given another 8-bit binary number, Y, and told that Y differs from X in exactly one bit position. How many bits of information about X have you been given?	6+4	CO2, CO3														



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Semester: July 2023-October 2023		
Maximum Marks: 30	Examination: In-Semester Examination	Duration: 1hr. 15 min.
Programme code: 03		
Programme: B.Tech Electronics and Communication Engineering	Class: FY/SY/TY/LY /MTECH	Semester: I/II/III/IV/V/VI/VII/VIII (SVU 2020)
Name of the Constituent College: J. Somaiya College of Engineering		Name of the department: COMP/ETRX/EXTC/IT/MECH
Course Code: 116U03C503	Name of the Course: Digital Signal Processing	

Question No.		Max. Marks
Q1	Compute 8-Point DFT using radix-2 DIT FFT of the given time signal, $x(n) = \{1, 2, 1, 2, 1, 3, 1, 3\}$. OR Calculate IDFT of the following frequency signal $X(K) = \{14, j1.414, 0, j1.414, -6, -j1.414, 0, -j1.414\}$	10
Q2	Attempt any Two (5 marks each) a) Compute circular convolution of the following sequences $x(n) = \{-1, 2, -2, -1\}$, $y(n) = \{1, -2, -1, -2\}$ b) Compute linear convolution of the following sequences $m(n) = \{-1, 2, -2, -1\}$, $h(n) = \{1, -2, -1, -2\}$ c) Plot the magnitude and phase spectrum of the following system and Describe the type of the filter. $H(z) = z^{-1}$	10
Q3	Realize the cascade and parallel structures of the system governed by the following difference equation, $y(n) - \frac{3}{10} y(n-1) - \frac{1}{10} y(n-2) = x(n) + \frac{1}{9} x(n-1)$	10



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VIDYAVIHAR UNIVERSITY

Semester: July 2023-October 2023

Maximum Marks: 30

Examination: In-Semester Examination

Duration: 1hr. 15 min.

Programme code: 03

**Programme: B.Tech Electronics and
Communication Engineering**

Class: TY

**Semester: V (SVU
2020)**

Name of the Constituent College:

J. Somaiya College of Engineering

Name of the department: EXTC

Course Code: 116U03E513

Name of the Course: Operating System

Question No.		Max. Marks																								
	Write short notes on (Any 4) :- 1) System calls. 2) Process Control Block. 3) Monolithic and Micro kernel . 4) What are the differences between user level and kernel level thread? 5) Discuss how deadlock could be detected. 6) Services of OS.	10																								
	Consider the following snapshot of the process to be executed. Draw the Gantt chart and determine the average waiting time and average turnaround time for FCFS, SJF (non-preemptive), and round robin(Preemptive) (time quantum=2) scheduling algorithm. <table><tr><th>Process</th><th>Arrival Time</th><th>Burst Time</th><th>Priority</th></tr><tr><td>P1</td><td>0</td><td>6</td><td>2</td></tr><tr><td>P2</td><td>2</td><td>5</td><td>1</td></tr><tr><td>P3</td><td>3</td><td>4</td><td>2</td></tr><tr><td>P4</td><td>4</td><td>3</td><td>3</td></tr><tr><td>P5</td><td>5</td><td>1</td><td>4</td></tr></table> Note: Assume lower number indicates higher priority	Process	Arrival Time	Burst Time	Priority	P1	0	6	2	P2	2	5	1	P3	3	4	2	P4	4	3	3	P5	5	1	4	10
Process	Arrival Time	Burst Time	Priority																							
P1	0	6	2																							
P2	2	5	1																							
P3	3	4	2																							
P4	4	3	3																							
P5	5	1	4																							
	What is semaphores? Give an implementation of bounded buffer producer consumer problem using semaphores. OR What is critical section of process? Explain two process Peterson's solution to achieve mutual exclusion of critical sections in an OS with proper code and example.	10																								



SOMAIYA
VIDYAVIHAR UNIVERSITY

Examination: In-Semester Examination

Duration : 1Hr 15 Min

Maximum Marks: 30

Programme code: 03

Programme: Electronics and Telecommunication Engineering
Name of the Constituent College:

Class: TY
BTech

Semester: V
(SVU 2020)

Name of the Constituent College: K. J. Somaiya Coll.

K. J. Somaiya College of Engineering
Course Code: 11GEC002

Name of the department:
EXTC

Course Code: 116U03C501

Name of the Course: Digital Communications

Question No.		Max. Marks														
Q1	A. Draw unipolar NRZ, polar NRZ, bipolar NRZ and Manchester encoding for following data sequence. 1 0 1 0 1 1 1 0 0 B. What is meant by the Gram Schmidt orthonormalization process? What is the purpose of this process in digital communications?	05 + 05														
Q2	A discrete memoryless source has alphabet of six symbols with probability for its output as given below: <table border="1"><tr><td>Symbol</td><td>S₀</td><td>S₁</td><td>S₂</td><td>S₃</td><td>S₄</td><td>S₅</td></tr><tr><td>Probability</td><td>0.1</td><td>0.05</td><td>0.3</td><td>0.15</td><td>0.2</td><td>0.2</td></tr></table> (i) Design the Huffman code for the above symbol (ii) Find Entropy (iii) Compute average codeword length (iv) Calculate Code Efficiency OR A. For a television transmission, the required number of brightness level is equal to 32, pixels /picture frame equal to 10⁶, frames transmitted per second equal to 30 and SNR equal to 40 dB. Find minimum bandwidth required. B. State and explain channel capacity theorem.	Symbol	S ₀	S ₁	S ₂	S ₃	S ₄	S ₅	Probability	0.1	0.05	0.3	0.15	0.2	0.2	10
Symbol	S ₀	S ₁	S ₂	S ₃	S ₄	S ₅										
Probability	0.1	0.05	0.3	0.15	0.2	0.2										
Q3	A. A waveform, $x(t) = 10 \cos (1000t + \pi/3) + 20 \cos (2000t + \pi/6)$ is to be uniformly sampled for digital transmission. What is maximum allowable time interval between sample values that will ensure perfect signal reproduction? B. Explain sampling theorem for lowpass signal with necessary diagrams. OR B. Explain effect of aliasing. How to avoid aliasing?	05														
		05														

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)
Semester: August-December 2022
In-Semester Examination

LY B. Tech
h: EXTC

Name of the course: Machine Learning
ion: 1hr.15 min

Semester: VII
Course Code: 2UTE722
Max. Marks: 30

Name of the course: Machine Learning		Max. Marks																			
Duration: 1hr.15 min																					
Questions		Marks																			
It is required to fit a regression model $y_i = x_i \beta_1 + \epsilon_i$ to the following data using Least Square method. Find the value of model parameter β_1. Plot the scatterplot of data together with the fitted regression model line. <table><tr><td>x</td><td>-2</td><td>0</td><td>3</td></tr><tr><td>y</td><td>1</td><td>2</td><td>4</td></tr></table> OR Explain the following with reference to Logistic regression algorithm for binary classification: a) Odds ratio b) Logit function c) Cost function d) Methods to obtain Logistic regression model parameters e) Suppose data consists of a single predictor and a response variable and a logistic regression model is fitted to the data. Provide interpretation of model parameters.		x	-2	0	3	y	1	2	4	10 marks											
x	-2	0	3																		
y	1	2	4																		
Consider three random variables P_1, P_2 and Y. P_1 and Y are generated according to Tables 1 and 2. Once P_1 has been generated, P_2 takes on the same value as P_1. Consider a Naive Bayes classifier trained on features P_1 and P_2 and class label Y. What will be the prediction of the classifier for $P_1 = 0, P_2 = 1$? Assume threshold of 0.5. <table><tr><th>Table 1</th><th>$P(Y)$</th></tr><tr><td>$Y=0$</td><td>0.7</td></tr><tr><td>$Y=1$</td><td>0.3</td></tr></table> <table><tr><th>Table 2</th><th colspan="2">$P(P_1 Y)$</th></tr><tr><td></td><th>$P_1=0$</th><th>$P_1=1$</th></tr><tr><td>$Y=0$</td><td>0.6</td><td>0.4</td></tr><tr><td>$Y=1$</td><td>0.4</td><td>0.6</td></tr></table>		Table 1	$P(Y)$	$Y=0$	0.7	$Y=1$	0.3	Table 2	$P(P_1 Y)$			$P_1=0$	$P_1=1$	$Y=0$	0.6	0.4	$Y=1$	0.4	0.6	5 marks	
Table 1	$P(Y)$																				
$Y=0$	0.7																				
$Y=1$	0.3																				
Table 2	$P(P_1 Y)$																				
	$P_1=0$	$P_1=1$																			
$Y=0$	0.6	0.4																			
$Y=1$	0.4	0.6																			
Explain in brief k-nearest neighbour algorithm with its merits and demerits.		5 marks																			

Q3A

Consider following dataset for classifying whether a type of cloth fabric is appealing or not.

Texture	Color	Price	Appealing
Rough	Dull	High	Yes
Rough	Bright	High	Yes
Rough	Dull	Medium	Yes
Soft	Bright	Low	No
Soft	Bright	Low	No
Soft	Bright	High	Yes
Rough	Dull	Medium	Yes
Rough	Bright	Low	No
Soft	Dull	Medium	No
Soft	Dull	Medium	No

Identify which predictor will be chosen as a root node of the decision tree using information gain measure? Write all the steps of your calculations.

Q3 B

How to understand if a decision tree is overfitting? Briefly describe the techniques to avoid overfitting a decision tree.

6 marks

4 marks

K. J. Somaiya College of Engineering, Mumbai-77

(Autonomous College Affiliated to University of Mumbai)

Semester: August-December 2022

In-Semester Examination

Branch: LY B. Tech

Branch: EXTC

Name of the course: Real Time Operating Systems

Duration: 1hr.15 min

Semester : VII

Course Code: 2UTE721

Max. Marks: 30

No	Questions	Marks
Q1	Enlist the types of programming models in Embedded Systems. Suggest suitable programming model for Automatic hand sanitizer dispenser machine and explain.	10 marks
Q2	Solve any two I. What are deadlocks? And how to prevent systems from it. II. Note down any five RTOS examples with its features. III. Enlist any five kernel services. IV. Explain types of device drivers.	10 marks
Q3	Apply Round robin algorithm for scheduling and answer the following. I. arrival time II. completion time III. burst time IV. turn around time V. how it is different than Lottery scheduling OR How interrupt handling is done in RTOS? Explain the below terms. I. Interrupt recovery time II. Interrupt latency	10 marks

Semester: August 2022-December 2022		
Maximum Marks: 30	Examination: In-Semester	Examination Duration :1H 15min
Programme code: 03		
Programme: Electronics and Telecommunication Engineering	Class: TY	Semester: V (SVU 2020)
Name of the Constituent College: J. Somaiya College of Engineering	Name of the department: EXTC	
Course Code: 116U03C503	Name of the Course: Digital Signal Processing	

Question No.		Max. Marks
Q. 1	Find the circular convolution of the $x(n)$ and $h(n)$ if $x(n) = \{1, 0, 2, 3\}$ and $h(n) = \{1, 2, 4\}$	05
Q. 2	Obtain the Direct form – I realization of the following LTI system, $y(n) = -0.75 y(n-1) + 0.5 y(n-2) + 0.25 y(n-3) + x(n) + 4 x(n-1) + 3 x(n-2)$ OR Realize the Direct form- II structure of the following system $H(z) = \frac{(z+5)}{(z+0.4)(z+0.5)(z+0.6)}$	05
Q. 3	Compute a) The 4-point DFT of following sequence $x(n) = \{1, 2, 1, 3\}$. b) Find $x(n)e^{-j2\pi n/4}$, using $x(n)$ given above. c) Find $X(K-1)$, Where $X(K)$ is the DFT of given above $x(n)$ OR Perform linear convolution of the following sequences by a) Overlap add method b) Overlap save method $x(n) = \{1, -1, 2, -2, 1, -1, 2, 1, -1, 2\}$ $h(n) = \{2, -1, 3\}$	06 02 02 05 + 05
Q. 4	Design a linear phase low pass FIR filter using rectangular window by taking 7 samples of window sequence and with a cutoff frequency $\omega_c = 0.2\pi$ rad/sample.	10